

SMRTTECH 4AI3 - Artificial Intelligence and Machine Learning

Lab 1 - Python Basics

Instructors: Siqu Zhao

Due: Midnight 11:59PM, day before your next lab session

Please read and follow these instructions:

- Submit your lab on Avenue to Learn (A2L) before the due date (midnight of the day before your next lab session). The submission will be closed after the deadline.
- Do your lab exercises on your own, i.e., do not copy. If two lab reports (or some parts of them) are found to be copies of each other, then BOTH shall receive 0 mark.
- You may use Google Colab or any other Python programming platform of your own choice to run your programs.
- Please read the questions carefully and use AI / machine learning libraries only when specially required as part of the question. Also add comments in your code where ever necessary.
- Submit your working python source files (or Jupyter notebooks) in zipped or source code format.

Exercises

1. **Write a python program to swap first and last elements in a list.**

Example: If the given list is ['orange', 'apple', 'banana'] , the output should be ['banana', 'apple', 'orange'].

2. **Write a python program to print today, yesterday, and tomorrow's date.**

Example: If today's date is 10 Sep 2025, the output should be

Today's date: 10 Sep 2025

Yesterday's date: 9 Sep 2025

Tomorrow's date: 11 Sep 2025

3. **Write a python function to take a list of numbers as input and sort it in descending order.**

Example:

Input: [31, 100, 2, 64, 87]

Output: [100, 87, 64, 31, 2]

Input: [1, 2, 3, 4, 5, 6, 7]

Output: [7, 6, 5, 4, 3, 2, 1]

4. **Write a python program to find most occurring characters and its count from a given string (using dictionary to hold your word list).**

Example:

Input string: 'I am enrolled in a McMaster degree program. There are a number of McMaster degree programs listed in top 100 degree programs of the world.'

Output:

Most occurring word: degree

Word Count: 3